FULL-PAGE OPTICAL MUSIC RECOGNITION OF JAZZ LEAD SHEETS

Anonymous Authors
Anonymous Affiliations
anonymous@ismir.net

ABSTRACT

Optical Music Recognition (OMR) of handwritten jazz lead sheets remains a challenging problem due to the variability of chord notation and the limited availability of annotated data. While recent work has shown promising results on pre-segmented staff-level lead sheet transcription, full-page handwritten lead sheet recognition has remained unexplored. We present the first full-page OMR system for handwritten jazz lead sheets, adapting the Sheet Music Transformer with a curriculum learning strategy that progressively trains the model on inputs of increasing length, from single staff up to complete pages. To evaluate the system, we construct a detect-and-concatenate pipeline as a full-page baseline and propose an evaluation methodology that can separate harmonic transcription performance from melodic transcription performance. Experiments show that our full-page system consistently outperforms a comparison system working on pre-segmented staves.

1. INTRODUCTION

Optical Music Recognition (OMR) systems have advanced rapidly in recent years, enabling reliable transcription of printed staff notation into symbolic formats that support useful downstream tasks such as automatic transposition, part extraction, score editing, and computational music analysis. These tools provide meaningful convenience for performers and educators by digitizing traditional sheet music workflows [1]. Handwritten jazz lead sheets, however, present a distinct challenge for OMR. Unlike fully notated classical scores, lead sheets combine staff notation with chord symbols written above the staff, requiring transcription of both melodic and harmonic information, while handwriting introduces substantial visual variability beyond the printed notation settings. When no editable digital version is available, even simple tasks such as generating transposed parts —essential for musicians playing transposing instruments such as saxophone or trumpet— become unnecessarily cumbersome. Thus, reliable OMR for jazz lead sheets would provide an immediate and practical benefit by converting the image into symbolic notation that can be edited, transposed, and redistributed.

Recently, Martinez-Sevilla et al. presented one of the first approaches to handwritten jazz lead sheet OMR by introducing a dataset for this task and adapting a transformer-based system to recognize pre-segmented staff-level images [2]. This study reports that chord-symbol recognition remains a challenge in the staff-level formulation of handwritten jazz lead sheet OMR, as most transcription errors in their qualitative analysis are related to chord symbols rather than melody. The main reason for this is that chords are handwritten in shorthand that varies across writers (e.g., `C△7`, `Cmaj7`, and `CM7` denote the same chord). These difficulties raise the question of whether chord-symbol transcription may benefit from moving beyond isolated staff-level inputs. As recent literature notes, segmenting a page into individual staves inherently destroys visual context, making it impossible for models to assess cross-staff interactions [3]. Furthermore, segmentation-based approaches suffer from error propagation, where segmentation mistakes cascade into the transcription stage. By bypassing this two-stage pipeline, full-page input provides richer global context than staff-level crops, preserving complex reading orders across multiple systems and capturing cross-staff interactions that are otherwise lost with staff-level input. This full-page approach might be especially beneficial in jazz lead sheets for capturing harmonic patterns.

In this study, we investigate full-page OMR for handwritten jazz lead sheets and its effect on chord-symbol transcription. [1] Our main contributions are:

- The presentation of the first **end-to-end full-page handwritten OMR system** for jazz lead sheets transcribing both melodic and harmonic content. We show that this approach —trained with a curriculum learning strategy— reliably outperforms a state-of-the-art staff-level transcription approach.
- The introduction of an **extended set of chord-oriented evaluation metrics** to enable a finer understanding of model capabilities and error types.

2. RELATED WORK

The field of Optical Music Recognition (OMR) studies computational methods for transcribing images of musical scores into machine-readable symbolic formats such as Humdrum, MusicXML, and MEI [1]. Early OMR was typically formulated as a multi-stage document analysis pipeline, following the steps of staff-line detection and removal, symbol segmentation, symbol classification, and musical reconstruction [4–6]. Within this framework, music

[1] Full code available at: `https://anonymous.4open.science/r/FullPageJazzOMR-30D6/`

symbol recognition was approached with detecting vertical lines, note head, and beams [7, 8]. This pipeline-based view was influential in early OMR, however, it faced robustness challenges for higher complexity tasks such as polyphonic input, handwritten scores, and other visually diverse sheet-music inputs [9].

With the rise of deep learning, OMR adopted object-detection frameworks from computer vision, adapting general detectors such as YOLO [10], SSD [11], and RetinaNet [12]. Noteworthy is the work of Pacha et al. presenting a deep-learning baseline for general music object detection using Faster R-CNN [13], showing that such systems can identify a broad set of symbol classes directly from score images [14]. Subsequent studies extended these ideas to specialized handwritten settings: de Vega et al. applied Mask R-CNN to handwritten four-part harmony exercises [15], while Paul et al. used an ensemble of transfer-learned CNNs for isolated handwritten music symbol classification [16]. Together, these works mark a transition from symbol classification to CNN-based deep learning while still treating OMR as an object-detection task.

A different deep-learning direction reformulates OMR not as object detection, but as direct transcription from score images to symbolic sequences. In this line of work, Calvo-Zaragoza et al. explored sequence-based OMR for printed monophonic music using Convolutional Recurrent Neural Networks (CRNNs), in which a CNN extracts visual features from the score image and an RNN predicts the corresponding sequence in a custom serialized sequence [17]. The models were trained with Connectionist Temporal Classification (CTC) loss, allowing staff images to be transcribed directly into symbol sequences without explicit information about the location of the symbols in the image [18]. This same CRNN–CTC formulation was later extended to handwritten music that improved symbol-level accuracy on handwritten notation benchmarks [19]. However, to address the limitations of CTC when dealing with simultaneous musical symbols, Baró et al. established a new baseline for handwritten Western music targeting handwritten scores using independent frame-wise dense layers and a Smooth L1-loss function to predict pitch and rhythm simultaneously [20]. For more complex polyphonic piano notation, Mayer et al. argue that reliable end-to-end OMR also depends on the consistency of the symbolic target, introducing Linearized MusicXML (LMX) as a stable textual representation for training and evaluation [21]. Similarly breaking the monophonic barrier, Rios-Vila et al. introduced the Sheet Music Transformer (SMT), a CNN-to-Transformer architecture that directly predicts Humdrum **kern encoding from score images, successfully handling complex polyphonic structures at the staff level [22]. The input data used by studies of the sequence-based OMR systems mentioned above are pre-segmented single-staff samples, which reduce layout complexity and shorten the prediction sequence. A notable exception is recent work on full-page recognition. Rios-Vila et al. extended the work of SMT to full-page recognition with curriculum training, presenting the first end-to-end OMR approach designed to handle complex polyphonic structures at the page level [3].

Beyond input granularity, OMR difficulty mostly depends on the notation target being transcribed. Jazz lead sheets recognition as a related yet distinct challenge is relatively underexplored. Unlike classical polyphonic scores in common Western notation, lead sheets combine a largely monophonic melodic staff with chord symbols, introducing an additional recognition target. Martinez-Sevilla et al. presented the first handwritten jazz lead sheets dataset and baseline system for this task by adapting the SMT architecture to transcribe pre-segmented staff images [2].

Building on these lines of work, our study focuses on the intersection of three challenges that have largely been treated separately in prior OMR research: handwritten notation, full-page transcription, and jazz lead sheet recognition.

3. METHODOLOGY

This section presents our methodology for full-page OMR of handwritten jazz lead sheets. Our study aims to answer two questions. First, can an end-to-end model directly transcribe a complete handwritten lead sheet that outperforms a model relying on pre-segmented staff images? Second, in the absence of prior work on full-page handwritten jazz lead sheet recognition, what is an appropriate methodology and metrics to capture different aspects of recognition performance compared to a baseline? To answer these questions, we first describe the dataset and data-cleaning steps, then introduce the detect-and-concatenate baseline and the proposed full-page models, both of which build from the same staff-level SMT checkpoint. Finally, we define the standard and chord-oriented metrics used to evaluate the systems.

3.1 Dataset

We use the JazzMus dataset [2], a collection of 163 unique jazz standards consisting of synthetic and handwritten scores aligned with Humdrum and MusicXML encodings. [2] The synthetic portion includes 326 full-page renderings generated using MuseScore with two font styles (Classical and MuseJazz). The handwritten portion includes 293 pages produced by jazz professionals and students who manually copied the synthetic lead sheets by hand. All multiple copies of the same piece are restricted to the training set, preventing leakage of identical musical content across splits. The dataset also provides bounding boxes for each individual staff on the full page. [3]

3.1.1 *Data cleaning*

The JazzMus annotations were converted from MusicXML to Humdrum, which introduced conversion artifacts. First, we correct chord symbols in 14 full-page samples that were incorrectly represented as comments rather than chord annotations, thus, not parsed as ground truth. Second, we fix

[2] In this study, all output is represented in Humdrum notation; see [23] for the Humdrum reference guide.

[3] For more information on dataset and the staff-level jazz lead sheet OMR system, see: `https://grfia.dlsi.ua.es/jazz-omr/`, access date: Apr 27, 2026

latex/baseline_flowchart_v5.pdf

Figure 1. Overview of the detect-and-concatenate baseline showing staff detection, transcription, and page-level reconstruction.

misaligned synthetic staff-level bounding boxes that reused crop coordinates from the handwritten subset. We contributed these fixes back to the dataset repository through pull requests. [4] We also observe some accidentals and slur markings errors in the annotations from manual copying of lead sheets. We retain these cases because they provide a realistic setting for evaluating model's robustness to handwritten input.

3.2 Models

We adopt the Sheet Music Transformer (SMT) [22] as the base model architecture for all experiments. SMT is an encoder–decoder architecture for image-to-sequence music transcription, using a ConvNeXt encoder to extract visual features and a Transformer decoder to generate sequences in Humdrum notation. Unlike previous approaches relying on CTC loss, the SMT decoder predicts the probability of each symbol conditioned on both the visual features and the previously generated tokens with teacher forcing applied during training. Following previous work [2], we adopt the medium word-level tokenization identified as the best-performing tokenization scheme.

3.2.1 Staff-Level Baseline

Because prior work on handwritten jazz lead sheet OMR is studied at the staff level rather than the page level, there is currently no established baseline for evaluating full-page handwritten jazz lead sheet. As our study introduces full-page handwritten jazz lead sheet transcription, we construct

[4] `https://huggingface.co/datasets/PRAIG/JAZZMUS`, access date: Feb 23, 2026.

a detect-and-concatenate baseline to serve as a comparison system for our proposed model. As illustrated in Figure 1, the pipeline first detects individual staves on the page, then transcribes each staff independently using the staff-level SMT recognizer, and finally concatenates all the staff predictions into a page-level Humdrum output. This baseline preserves the staff-level recognition setup of earlier work while extending it to full-page input. It therefore provides a natural comparison system for assessing whether page-level training improves performance beyond what can be achieved by staff-level recognition alone.

Staff Segmentation: Page images are first preprocessed with resizing and de-skewing to normalize scale and correct page rotation. To perform the staff segmentation, we adopt YOLOv11s [24], an object detection model fine-tuned specifically for the music staff-detection task. However, we identified three problems with the raw bounding-box detections: missed staff detections, duplicate detections of the same staff, and bounding boxes that are too narrow vertically to capture chord symbols above the staff. We address these problems through the following post-processing steps. First, missed staff detections are recovered by analyzing the spacing between consecutive boxes: if the space between adjacent staffs exceeds the median staff distance for this page, a virtual bounding box with median dimensions is inserted. Second, YOLO occasionally predicts two overlapping boxes for the same staff. We identify these cases using one-dimensional vertical Intersection over Union (IoU) [25] and remove the smaller box. Third, the inter-staff gap of bounding-box detections is filled by assigning 25% of the gap to the upper crop and 75% to the lower crop. This expands each lower staff crop upward so that chord symbols written above its staff lines are included, while keeping adjacent crops non-overlapping.

Staff-Level Recognition: After the segmentation process, each cropped staff image is fed to the SMT model pretrained on the staff level study [2]. The staff predictions are concatenated into a full-page output: the first staff's header and spine declarations are preserved, subsequent staves have their Humdrum headers stripped, and all staves are joined with line-break notation in Humdrum (`!!linebreak:original`) separators, with spine terminators appended at the end. We then evaluate the concatenated results against the full-page ground truth.

3.2.2 Proposed Model

To support full-page transcription, we extend the SMT's output vocabulary with a dedicated `<linebreak>` token so that the model can explicitly represent staff boundaries and generate multi-staff outputs. This increases the vocabulary size from 154 to 155. Accordingly, the input and output of the decoder embedding layer are increased by one. In both layers, the staff-level pretrained weights, used in the baseline parameters, are preserved, while the parameters associated with the new `<linebreak>` token are initialized randomly. Beyond this change to the input embedding and output layer, the model architecture remains unchanged. The main methodological challenge is therefore not redesigning the model architecture itself, but bridging the gap between staff-level pretraining and full-page recognition. To address this, we investigate two curriculum learning strategies that progressively expose the model to longer and more structurally complex inputs.

Masking Curriculum: The masking curriculum achieves the transition to full-page transcription through vertical masking, which reveals larger top portions of each page while masking out the lower content. We apply a 9-stage curriculum learning strategy. At each stage $k \in \{1, \ldots, 9\}$, the training data are modified to reveal exactly k staves. This allows the model to gradually adapt to larger image and annotation inputs and longer output sequences as the stages progress.

The masking curriculum data is constructed from JazzMus full-page scans. For each full-page image, we generate a set of top-N crops by retaining only the top N staves and cropping away the remainder. For example, if a lead sheet page contains 8 staves, we generate 8 masked versions of that page, each retaining the top N systems for N = 1, …, 8. For each image crop, the ground truth annotation is truncated to the first N staves, deleting all subsequent symbols, and ending with the final Humdrum spine terminator. Although JazzMus full-page scans contain up to 11 staves, we restrict the masking curriculum to 9 stages. This choice results from the staff-count distribution of the full-page subset: the majority of pages contain between 6 and 8 staves (207 of 294 pages), whereas only 28 pages contain 9 or more. Treating 9-, 10-, and 11-staff pages as separate stages would therefore add little useful variation, since these pages already function as near-complete or complete full-page inputs while contributing only a small number of additional training examples. Thus, we group all pages with 9 or more staves into stage 9, which serves as the final full-page stage. This data preparation procedure yields 1,688 training samples and 105 validation samples. At each curriculum stage, the training subset contains 245 handwritten and 278 synthetic samples, while the validation subset contains 16 samples. The test set is used only at the final stage and consists of 32 unmodified full-page samples.

Training begins from the pretrained staff-level SMT checkpoint and proceeds through the curriculum schedule using a staged advancement strategy. At stage k, the training and validation set include only samples with k staves. Once a sample reaches its maximum available number of staves, it remains unchanged as a full-page sample in all subsequent stages. The stage advances once the validation metrics (see 3.3) falls below 20% for 3 consecutive validation epochs without improvements. The 20% threshold serves as a readiness check, since preliminary runs showed that stages typically converged around 15%. At the final stage ($k = 9$), all samples are full-page without any masking and training continues until early stopping with a patience of 10 validation checks.

A direct consequence of this training strategy is that when the stage shifts from $k-1$ to k, samples from earlier stages disappear from the training sample entirely, and the model risks forgetting shorter-input behavior it previously learned. To monitor this, after each stage we perform a full validation sweep over all nine validation subsets. Preliminary runs without replay showed that models trained on later stages degraded substantially on shorter-input subsets, indicating forgetting of earlier-stage behavior. To mitigate this, we introduce experience replay: at each stage, a fraction r of samples from earlier stages is retained in the training pool, where r denotes the replay ratio. We study four replay settings, with replay ratios $r \in \{0, 0.25, 0.50, 1.00\}$, to examine how retaining earlier-stage samples affects stability and final performance.

Stacking Curriculum: The stacking curriculum follows the same 9-stage curriculum training procedure as the masking approach, differing only in the way the training data are prepared. Rather than masking full-page scans, we construct longer input samples by synthesizing multi-staff examples from individual staff crops. At stage k, samples are stacked by selecting one staff crop and sampling $k - 1$ additional staves from other examples. The selected staves are resized to a uniform height with aspect ratio preserved, padded to a common width, and vertically concatenated into a single multi-staff image resembling a full-page lead sheet. The corresponding ground-truth sequence is formed by concatenating the per-staff annotations in stacking order. The annotation of the first staff is kept with its original Humdrum header and spine declarations, while subsequent staff annotations have their headers removed and are joined using the <linebreak> token.

At stage $k = 1$, the data consist of the training set of 1,688 handwritten and 1,872 synthetic staff-level entries, and a validation set of 105 handwritten samples, which is all individual staves derived from the training and validation set. For stages $k = 2, \ldots, 9$, the same number of training and validation is sampled using the stacking method. At the final stage (k = 9), 245 handwritten and 278 synthetic real full-page images are additionally appended to the training pool so that the model is exposed to authentic page-level layout before evaluation. This results in a total of 32,563 training samples and 945 validation samples across the curriculum. The test set consists of 32 unmodified full-page samples, identical to the masking curriculum. Because the stacking curriculum relies on synthetically generated training and validation data, the number of samples produced at each stage is itself a design choice. Here, we set this number to match the total number of available staff-level examples to remain consistent with the pretrained staff-level checkpoint. A potential future work may study how different sample counts influence training dynamics and final performance. A practical drawback of increasing training size is that the larger number of generated samples in the stacking curriculum significantly increases the computational cost, both in memory usage and in training time.

Training begins from the same pretrained staff-level SMT checkpoint and proceeds through the stacking curriculum using the same staged advancement and final stage early-stopping strategy as in the masking setting. To account for increased training time due to the larger number of training samples, experience replay is applied with a fixed ratio of $r = 0.25$, balancing training time and not forgetting earlier stage training.

Experimental Setup: Both curricula are optimized with AdamW using 150 warmup steps, with validation performed every 10 epochs. For the masking curriculum, stages 1–8 use a batch size of 8 and a learning rate of 5×10^{-5}, while stage 9 uses a batch size of 1 and a learning rate of 5×10^{-6} due to increased memory requirements. For the stacking curriculum, stages 1–5 are trained with a batch size of 8 and a learning rate of 5×10^{-5}. Stages 6–8 use a batch size of 4 and a learning rate of 1×10^{-5}, while stage 9 uses a batch size of 1 and a learning rate of 5×10^{-6}. All experiments are conducted on a single NVIDIA H200 GPU using BF16 mixed precision.

3.3 Evaluation Metrics

Following prior work in OMR and end-to-end music transcription [1–3,22], we report three edit-distance-based error rates as overall measures of model performance: Character Error Rate (CER), Symbol Error Rate (SER), and Line Error Rate (LER), computed as the Levenshtein edit distance between the prediction and ground truth, divided by the total number of ground-truth units at the corresponding level and multiplied by 100. CER is computed at the character level and indicates individual character errors. SER is computed at the symbol level, assessing the correctness of individual musical symbols. LER is computed at the line level. In our setting, LER implicitly evaluates line-level structural errors, such as assigning a chord symbol to the wrong note line. However, both SER and LER may still treat partially correct predictions as a whole edit-distance substitution error, making them unable to distinguish between minor and major mismatches within their respective units. To isolate melodic accuracy, the metrics are also computed on the melodic spine only. These melody-only metrics are denoted as: $\mathrm{CER}_{\mathrm{Melo}}$, $\mathrm{SER}_{\mathrm{Melo}}$, and $\mathrm{LER}_{\mathrm{Melo}}$.

While the traditional OMR metrics provide an overview of transcription accuracy, they are insufficient for isolating harmonic performance as melodic and harmonic errors are treated equally. To allow for differentiation between melodic and harmonic recognition performance, we introduce two chord-oriented metrics based on extracting chord symbols from the **mxhm spine into an ordered list and disregarding the spacing between them. The two proposed metrics are defined as:

- SER_{Root}: the Levenshtein distance over the number of chord symbols in the ground truth between predicted and ground truth chord sequences which are reduced to their root-note tokens only. This metric measures errors in chord-root recognition independently of the remaining chord specification.
- SER_{Chord}: the Levenshtein distance over the number of chord symbols in the ground truth between full predicted and ground truth chord sequences. Under this metric, a chord is counted as correct only when its complete symbolic form is transcribed correctly, including root, quality, extensions, and inversion.

These two metrics are intended to capture different lev-

Method	CER	SER	LER
Baseline	13.55	12.78	32.35
Masking, $r = 0$	17.61	16.80	31.41
Masking, $r = 0.25$	17.48	16.73	29.13
Masking, $r = 0.50$	14.11	13.59	26.25
Masking, $r = 1.00$	**13.32**	**12.50**	**23.15***
Stacking, $r = 0.25$	24.30	22.90	44.55

Table 1. Overall full-page recognition results on test set. Stars indicate statistical significance over baseline. (* $p < 0.01$)

els of harmonic correctness. In jazz lead sheets, the chord root conveys harmonic progression and leads the song structure, so root-note errors are often more consequential than mistakes in extensions or other symbolic details. SER_{Root} therefore evaluates whether the model recovers the harmonic backbone of the piece, while SER_{Chord} provides a stricter measure that also requires correct transcription of chord quality, extensions, and inversion. Comparing the two helps distinguish between errors that change the underlying harmony and errors that miss finer chordal detail.

4. RESULTS

We present the results in two parts. First, we compare the full-page curriculum models with the baseline using standard OMR edit-distance metrics. Second, we report melody-spine metrics and chord-oriented harmonic metrics to determine where the improvements occur.

4.1 Overall Full-Page Recognition

The overall full-page recognition results are shown in Table 1. This result matches the previous staff-level study with minor increase within $\pm 2 - 3\%$. Among the masking curriculum models, performance improves consistently as the replay ratio increases. Without replay ($r = 0$), the model performs worse than the baseline in CER and SER, although its LER is slightly lower than the baseline. A small amount of replay ($r = 0.25$) produces only minor improvement, while ($r = 0.50$) brings CER and SER much closer to the baseline and further reduces LER. The best-performing model is the masking curriculum with full replay ($r = 1.00$), which slightly improves over the baseline in CER and SER. A paired test shows only LER improves significantly ($p < 0.01$). The consistent LER improvement suggests that full-page training helps the model produce better line-level accuracy. Since LER is the only standard metric that implicitly reflects chord–melody alignment, this improvement may indicate better alignment of chord symbols. In contrast, CER and SER improve only when sufficient replay is used. One possible explanation is that higher replay ratios substantially increase the amount and diversity of training data by retaining more prior-stage samples, which supports stronger character- and symbol-level performance in later curriculum stages.

Nevertheless, the stacking curriculum performs worse than both the baseline and the masking curriculum on the overall metrics. Although the stacking method provides more training data, the number of real full-page samples is relatively small compared to the large amount of stacked data. This suggest that combining staves from different full-page samples may produce layouts that differ too much from real full-page examples for effective generalization. Even when preserving each staff's aspect ratio to avoid stretching, the stacked samples still exhibit inconsistent visual staff lengths, different numbers of bars, and discontinuities in key and chord progressions across staves. We experimented fine-tuning the final stacking method checkpoint on only the complete full page training sample, but the result improved only marginally. The limited improvement from fine-tuning suggests that simply exposing the model to real full-page samples after stacking curriculum training is not sufficient to overcome the mismatch introduced during earlier stacking training.

Metric	Baseline	Masking, $r = 1.00$	Δ
$\mathrm{CER}_{\mathrm{Melo}}$	11.96	**11.63**	-0.33
$\mathrm{SER}_{\mathrm{Melo}}$	14.40	**13.02**	-1.38
$\mathrm{LER}_{\mathrm{Melo}}$	26.66	**18.67***	-7.99
$\mathrm{SER}_{\mathrm{Root}}$	29.17	**21.25***	-7.92
$\mathrm{SER}_{\mathrm{Chord}}$	47.82	**30.67****	-17.15

Table 2. Spine-specific metrics across masking replay ratios. Stars indicate statistical significance over baseline. (* $p < 0.01$, ** $p < 0.001$)

4.2 Melodic and Harmonic Recognition

To separate melodic and harmonic transcription performance, we show spine-specific metrics on the melodic **kern spine and our proposed chord symbol metrics in Table 2. The masking curriculum shows a clear benefit from experience replay following the same pattern in the overall results. These results suggest that full-page training is especially beneficial for chord-symbol recognition. A likely reason is that the staff-level baseline observes only a short harmonic segment at a time, with an average of 5.78 chords per staff, whereas the full-page model observes average of 39.89 chords per page. This broader harmonic context may help the model learn more likely chord progressions, leading to more accurate chord transcription across the page.

Pair	Baseline	Masking, r=1.00
C $\leftrightarrow$ G	42	18
B $\leftrightarrow$ D	27	2
B $\leftrightarrow$ E	24	10
B♭ $\leftrightarrow$ E♭	19	7
A $\leftrightarrow$ D	15	6

Table 3. Top-5 most frequent root-letter confusion pairs in baseline and masking model r=1.00

latex/per_n_results_v2.pdf

Figure 2. Error rates as a function of staves per page N, for baseline and masking model (r=1.00). Numbers below each report the number of test pages.

5. DISCUSSION

To better understand why full-page training improves chord recognition, we analyze model behavior through three questions: what types of chord errors are reduced, where on the page the improvements occur, and when full-page context provides the greatest advantage.

First, we investigate the most frequency root-letter confusions; the five most common confusion pairs in the baseline are listed in Table 3. Unsurprisingly, all of them can be visually ambiguous in handwriting. All five drop substantially for the full-page model, as can be expected given the overall result.

We also examine chord errors by distance from the nearest <linebreak> token to test whether improvements were concentrated near staff boundaries. It is hypothesized that full-page training could preserve harmonic continuity across adjacent staves, while staff-level crops may disrupt this context. However, the reduction in chord errors is similar across distances, with an overall reduction of 48.1%. This suggests that full-page training improves chord recognition more broadly across the page, rather than only at system boundaries.

How the two systems behave on the number of staves per test page N provides another way to test the role of full-page context. Figure 2 plots $\mathrm{SER}_{\mathrm{Melo}}$ and $\mathrm{SER}_{\mathrm{Root}}$ against N for both models. On denser pages ($N \geq 7$), which account for the majority of the test set, the full-page model outperforms the baseline on chord recognition. The baseline remains competitive with the full-page model in melodic accuracy overall and in chord recognition when the number of staves is small. This suggests that, as pages contain more staves and more harmonic context, the full-page model's access to global context translates into a larger advantage. However, this trend may also reflect an uneven distribution of page size in the training and test set.

6. CONCLUSION

We have presented the first full-page handwritten OMR system for jazz lead sheets, comparing a detect-and-concatenate baseline with two curriculum learning approaches that train the model on inputs of increasing length. The masking curriculum with full experience replay outperforms the baseline on all evaluation metrics. The proposed chord-specific metrics, $\mathrm{SER}_{\mathrm{Root}}$ and $\mathrm{SER}_{\mathrm{Chord}}$, show that out page-level system is beneficial for chord symbol transcription, specifically on lead sheets with a high number of staves.

Future work may build on the data-cleaning process described in Section 3.1.1 by further reducing minor transcription inconsistencies in the annotations. Data scarcity is also an important challenge, and existing kern-encoded jazz resources, such as the Charlie Parker Omnibook transcriptions used in [26], may provide a useful starting point for building larger handwritten lead-sheet datasets. Transfer learning from large vision-language models may also offer a possible direction for reducing dependence on OMR-specific pretraining annotations.

7. REFERENCES

[1] J. Calvo-Zaragoza, J. H. Jr., and A. Pacha, "Understanding optical music recognition," *ACM Comput. Surv.*, vol. 53, no. 4, Jul. 2020.

[2] J. C. Martinez-Sevilla, F. Foscarin, P. Garcia-Iasci, D. Rizo, J. Calvo-Zaragoza, and G. Widmer, "Optical music recognition of jazz lead sheets," in *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*, 2025.

[3] A. Ríos-Vila, J. Calvo-Zaragoza, D. Rizo, and T. Paquet, "End-to-end full-page optical music recognition for pianoform sheet music," *International Journal of Computer Vision*, vol. 134, no. 2, p. 49, 2026.

[4] D. Blostein and H. S. Baird, *A Critical Survey of Music Image Analysis*. Berlin, Heidelberg: Springer Berlin Heidelberg, 1992, pp. 405–434. [Online]. Available: https://doi.org/10.1007/978-3-642-77281-8_19

[5] D. Bainbridge, "Extensible optical music recognition," Doctor of Philosophy thesis, University of Canterbury, Christchurch, New Zealand, 1997. [Online]. Available: http://hdl.handle.net/10092/9420

[6] D. Bainbridge and T. Bell, "The challenge of optical music recognition," *Computers and the Humanities*, vol. 35, no. 2, pp. 95–121, 2001.

[7] A. Rebelo, G. Capela, and J. S. Cardoso, "Optical recognition of music symbols: A comparative study," *International Journal on Document Analysis and Recognition (IJDAR)*, vol. 13, no. 1, p. 19–31, Mar. 2010.

[8] A. Baró, P. Riba, and A. Fornés, “Towards the recognition of compound music notes in handwritten music scores,” in *2016 15th International Conference on Frontiers in Handwriting Recognition (ICFHR)*, 2016, pp. 465–470.

[9] A. Hartelt and F. Puppe, “Optical medieval music recognition using background knowledge,” *Algorithms*, vol. 15, no. 7, 2022. [Online]. Available: https://www.mdpi.com/1999-4893/15/7/221

[10] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 779–788.

[11] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, “Ssd: Single shot multibox detector,” in *Computer Vision – ECCV 2016*, B. Leibe, J. Matas, N. Sebe, and M. Welling, Eds. Cham: Springer International Publishing, 2016, pp. 21–37.

[12] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 42, no. 2, pp. 318–327, 2020.

[13] S. Ren, K. He, R. Girshick, and J. Sun, “Faster r-cnn: towards real-time object detection with region proposal networks,” in *Proceedings of the 29th International Conference on Neural Information Processing Systems - Volume 1*, ser. NIPS’15. Cambridge, MA, USA: MIT Press, 2015, p. 91–99.

[14] A. Pacha, J. Hajič, and J. Calvo-Zaragoza, “A baseline for general music object detection with deep learning,” *Applied Sciences*, vol. 8, no. 9, 2018. [Online]. Available: https://www.mdpi.com/2076-3417/8/9/1488

[15] F. F. De Vega, J. Alvarado, and J. V. Cortez, “Optical music recognition and deep learning: An application to 4-part harmony,” in *2022 IEEE Congress on Evolutionary Computation (CEC)*. IEEE Press, 2022, p. 01–07.

[16] A. Paul, R. Pramanik, S. Malakar, and R. Sarkar, “An ensemble of deep transfer learning models for handwritten music symbol recognition,” *Neural Computing and Applications*, vol. 34, no. 13, p. 10409–10427, jul 2022.

[17] J. Calvo-Zaragoza and D. Rizo, “End-to-end neural optical music recognition of monophonic scores,” *Applied Sciences*, vol. 8, no. 4, 2018. [Online]. Available: https://www.mdpi.com/2076-3417/8/4/606

[18] M. Alfaro-Contreras, J. M. Iñesta, and J. Calvo-Zaragoza, “Optical music recognition for homophonic scores with neural networks and synthetic music generation,” *International Journal of Multimedia Information Retrieval*, vol. 12, no. 1, p. 12, 2023. [Online]. Available: https://doi.org/10.1007/s13735-023-00278-5

[19] J. Calvo-Zaragoza, A. H. Toselli, and E. Vidal, “Handwritten music recognition for mensural notation with convolutional recurrent neural networks,” *Pattern Recognition Letters*, vol. 128, pp. 115–121, 2019.

[20] A. Baró, P. Riba, J. Calvo-Zaragoza, and A. Fornés, “From optical music recognition to handwritten music recognition: A baseline,” *Pattern Recognition Letters*, vol. 123, pp. 1–8, 2019.

[21] J. Mayer, M. Straka, J. Hajič, and P. Pecina, “Practical end-to-end optical music recognition fornbsp;pianoform music,” in *Document Analysis and Recognition - ICDAR 2024: 18th International Conference, Athens, Greece, August 30–September 4, 2024, Proceedings, Part VI*. Berlin, Heidelberg: Springer-Verlag, 2024, p. 55–73.

[22] A. Ríos-Vila, J. Calvo-Zaragoza, and T. Paquet, “Sheet music transformer: End-to-end optical music recognition beyond monophonic transcription,” in *Document Analysis and Recognition - ICDAR 2024: 18th International Conference, Athens, Greece, August 30–September 4, 2024, Proceedings, Part VI*. Berlin, Heidelberg: Springer-Verlag, 2024, p. 20–37.

[23] D. Huron, *Music Research Using Humdrum: A User’s Guide*. Stanford, CA: Center for Computer Assisted Research in the Humanities, 1999. [Online]. Available: https://www.humdrum.org/guide/

[24] J. V. Dvorák, J. Hajič and J. Mayer, “Staff layout analysis using the yolo platform,” in *Proceedings of the 6th International Workshop on Reading Music Systems*, 2024, p. 18.

[25] J. Yu, Y. Jiang, Z. Wang, Z. Cao, and T. Huang, “Unitbox: An advanced object detection network,” in *Proceedings of the 24th ACM International Conference on Multimedia*, ser. MM ’16. New York, NY, USA: Association for Computing Machinery, 2016, p. 516–520.

[26] D. Baker, A. Rosado, D. Shanahan, and E. Shanahan, “The role of idiomaticism and affordances in bebop improvisation,” in *Proceedings of the 14th International Conference on Music Perception and Cognition (ICMPC)*, 2016.